

Strategic Analysis of NYC Citi Bike Service 2022

A Data-Driven Deep Dive into Urban Mobility Imbalances and
Operational Optimization

Case Study: Dr. Lylan Bramasole
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The Challenge: Network Imbalance

Recurring Service Bottlenecks

Citi Bike faces a critical operational challenge: high-demand stations frequently encounter "dock-outs" (empty) or full docks during peak hours.

These imbalances disrupt the user experience, particularly for commuters, and necessitate high-cost manual fleet rebalancing via vans and valet services.



Integrated Data Foundation

By enriching 20 million trip records with environmental metrics, we identify the primary drivers of demand variance.

Data Source	Metric Categories	Volume / Granularity
Citi Bike Open Data	Timestamps, GPS, Station Names, Rider Type	20,000,000+ Records (2022)
NOAA Weather API	Temperature (TAVG), Precipitation, Wind Speed	Daily Averages (LaGuardia Station)
Engineered Data	Route IDs, Duration Clusters, Peak Hour Tiers	Cleaned Python Aggregations

Methodology: The Analytical Pipeline



Data Ingestion

Concatenating 12 monthly CSV files via AWS S3 to handle large-scale mobility data seamlessly.



Preprocessing

Cleaning GPS anomalies and outliers while standardizing datetime formats for cross-API merging.



Integration

Merging high-volume trip records with NOAA weather data using optimized key-value mapping.



Analysis

Spatial clustering and seasonal trend modeling to isolate the



Deployment

Developing an interactive Streamlit dashboard for



Optimization

Final algorithmic adjustment for rebalancing priorities based on

Results & Insights

Uncovering the Spatiotemporal Dynamics of NYC
Cycling

Station Demand Concentration

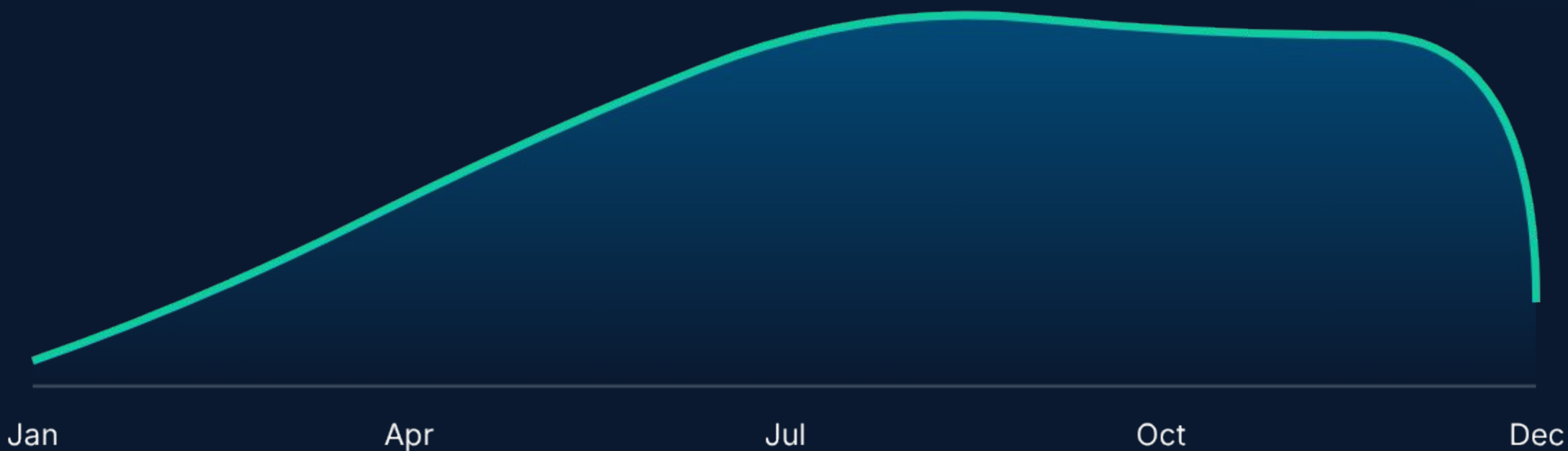
Top 5 Stations by Departure Volume. Demand is heavily skewed toward transit hubs and business districts.



Ridership Peaks with Temperature

**Peak Volume: ~3.8M Rides in
September**

Daily analysis confirms a direct linear correlation between Average Temperature and trip volume.



Route Dynamics: Commuter vs. Leisure

Weekday Patterns

The system acts as a commuter backbone. Flows are highly directional: Residential areas to Business Districts (AM) and the inverse during (PM) rush hours.

Imbalance is most acute at 8-9 AM and 5-6 PM.

Weekend Patterns

Activity shifts toward leisure loops. High concentration around Central Park, Hudson River Greenway, and waterfront paths in Brooklyn.

Routes are often non-linear, beginning and ending at the same station, suggesting round-trip recreational use.

| The Extreme Long-Tail Distribution

75%

of unique routes occur only once

Operational Efficiency Focus: Despite millions of trips, the majority of unique point-to-point connections are rare events.

Strategy must prioritize the high-frequency corridors that account for the bulk of ridership. Efficiency is won or lost in the top 1% of routes.

| Geographic Gaps

Outer Borough Opportunities

While Manhattan is saturated with stations, significant portions of Queens, Brooklyn, and the Bronx remain "service deserts."

Scaling dock density in these high-potential residential zones can bridge the gap between leisure riding and daily utility transit for underserved communities.



Strategic Recommendations



Optimize Rebalancing

Target manual redistribution to the top 20 high-frequency commuter corridors to prevent dock-outs during AM/PM peaks.



Seasonal Maintenance

Utilize the Jan-Feb ridership dip for heavy fleet overhaul, ensuring maximum bike availability for the spring surge.



Data-Led Expansion

Prioritize new station density in transit-dense outer borough corridors to convert leisure riders into daily utility commuters.

Questions?

Thank you for your attention to this strategic analysis.

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nyccitibike-servicestrategicanalysis.streamlit.app



https://github.com/L-Bramasole/NYC_CitiBike-service_Strategic_Analysis



[Dr. Lylan Bramasole | Data Analyst](#)

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